Palmer Penguins Analysis

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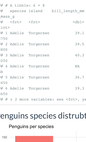
Exploratory Data Analysis

View data

```
library(palmerpenguins)

##
## Attaching package: 'palmerpenguins'

##
## The following objects are masked from 'package:datasets':
##
##     penguins, penguins_raw

data(penguins)
head(penguins)

##  species island    bill_length_mm bill_depth_mm flipper_length_mm body
##  <fct>   <fct>              <dbl>        <dbl>             <int> <int>
##  1 Adelie  Torgersen           39.1         18.7               181  3750
##  2 Adelie  Torgersen           39.5         17.4               186  3800
##  3 Adelie  Torgersen           40.3         18                 195  3250
##  4 Adelie  Torgersen           NA           NA                  NA    NA
##  5 Adelie  Torgersen           36.7         19.3               193  3450
##  6 Adelie  Torgersen           39.3         20.6               190  3650
##  # ... 2 more variables: sex <fct>, year <int>
```

Penguins species distrubition

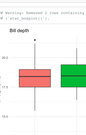



```
##
## Attaching package: 'dplyr'

##
## The following objects are masked from 'package:stats':
##
##     filter, lag

##
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

ggplot(penguins, aes.x = species, y = body_mass_g, fill = species) + geom_b

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

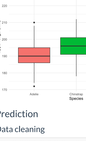
```
colSums(is.na(penguins))

##       species        island  bill_length_mm  bill_depth_mm
##             0             0               2              2
##ff_flipper_length_mm     body_mass_g       sex          year
##             2             2              11              0

ggplot(penguins, aes.x = island, fill = species)+ geom_bar() + labs(title
```

```
ggplot(penguins, aes.x = species, y = bill_length_mm, fill = species) + geo

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
ggplot(penguins, aes.x = species, y = bill_length_mm, fill = species) + geom

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



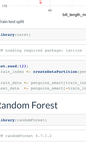

```
ggplot(penguins, aes.x = species, y = flipper_length_mm, fill = species) +

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Prediction

Data cleaning

```
penguins_omit <- na.omit(penguins)

ggplot(penguins_omit, aes.x = bill_length_mm, y = flipper_length_mm, color
= species, shape = island, size = 1.5)+
  labs(title = "Relation between length of bill to length of flipper",
       subtitle = "Visible separation between species", p
  theme_minimal()

train_data <- penguins_omit[train_index, ]
test_data <- penguins_omit[-train_index, ]
```



Train-test split

```
library(caret)

## loading required package: lattice

set.seed(123)
train_index <- createDataPartition(penguins_omit$species, p = 0.8, list =

train_data <- penguins_omit[train_index, ]
test_data <- penguins_omit[-train_index, ]
```

Random Forest

```
library(randomforest)

## randomforest 4.7-1.2

## Type 1(Bonus) to see new features/changes/bug fixes.

##
## Attaching package: 'randomforest'

## The following object is masked from 'package:dplyr':
##
##     combine

## The following object is masked from 'package:ggplot2':
##
##     margin

rf_model <- randomforest(species ~ bill_length_mm + bill_depth_mm + flipper

print(rf_model)

##
##Call:
## randomforest(formula = species ~ bill_length_mm + bill_depth_mm +
##               body_mass_g + island, data = train_data,          importan
## ce=TRUE)
##               Type of random forest: classification
##                     Number of trees: 500
## No. of variables tried at each split: 2
##
##         OOB estimate of  error rate: 1.49%
## Confusion matrix:
##        Adelie Chinstrap Gentoo class.error
## Adelie    115         2     0  0.01704602
## Chinstrap   2        63     1  0.04839346
## Gentoo      1         0   119  0.00800000

predictions <- predict(rf_model, test_data)

accuracy <- mean(predictions == test_data$species)

print(accuracy)

## [1] 0.9661154

table(Predicted = predictions, Actual = test_data$species)

##           Actual
## Predicted  Adelie Chinstrap Gentoo
##   Adelie      28         0     0
##   Chinstrap    0        12     0
##   Gentoo       0         0    29

varImpPlot(rf_model)
```

Conclusion

The Palmer Penguins dataset consists of various morphological parameters, including bill length, bill depth, flipper length, and body mass. The Random Forest model proved to be highly effective, achieving an accuracy of approximately 96.66% in classifying the three species.

Analysis of variable importance indicates that bill length, bill depth, and island are the most critical predictors for species differentiation. Furthermore, body mass_g was found to be the least influential variable. This result is consistent with the exploratory data analysis, which shows a significant overlap in body mass between the Adelie and Chinstrap species, making it a less reliable metric for classification.

Prepared by bill dimensions.